

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA0606

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

Case Study

Code of Conduct:

*I SARSHINI.S – 192321113 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

SARSHINI

Annexure A

Short Answer Test

1. A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.

Intelligent Disease Diagnosis System using Bayesian Methods

Aim

To design a probabilistic model for disease diagnosis using Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, and to analyze how uncertainty and conditional dependencies influence predictions.

Introduction

In medical diagnosis, uncertainty is inherent due to incomplete patient information, overlapping symptoms, and variability in disease progression. Probabilistic models such as Bayes' Theorem and Bayesian networks provide a systematic approach to handle uncertainty and improve decision-making.

Bayes' Theorem for Diagnosis

Bayes' Theorem is used to compute the probability of a disease given observed symptoms:

Where:

- $P(\text{Disease} \mid \text{Symptoms}) \rightarrow$ Posterior probability
- $P(\text{Symptoms} \mid \text{Disease}) \rightarrow$ Likelihood
- $P(\text{Disease}) \rightarrow$ Prior probability
- $P(\text{Symptoms}) \rightarrow$ Evidence

Application:

Given symptoms like fever, cough, and fatigue, the system calculates the probability of diseases such as flu, COVID-19, or pneumonia.

Naïve Bayes Classifier

Model Design

- Assumes all symptoms are conditionally independent given the disease
- Computes probability for each disease class

Formula:

$$P(D \mid S_1, S_2, \dots, S_n) \propto P(D) \times \prod P(S_i \mid D)$$

Steps:

1. Collect patient symptoms and medical history
2. Calculate prior probabilities of diseases
3. Compute likelihood of symptoms for each disease
4. Predict disease with highest posterior probability

Advantage:

- Simple and fast
- Works well with large datasets

Limitation:

- Assumes independence among symptoms (often unrealistic)

Bayesian Belief Networks (BBN)

Model Structure

- Represented as a **Directed Acyclic Graph (DAG)**
- Nodes: Diseases, Symptoms, Risk Factors
- Edges: Conditional dependencies

Example:

- Smoking $\rightarrow$ Lung Disease
- Lung Disease $\rightarrow$ Cough
- Fever $\rightarrow$ Infection

Key Features:

- Captures dependencies between symptoms
- Allows reasoning under uncertainty
- Supports inference even with missing data

Handling Uncertainty

1. **Probabilistic Representation:**
 - Uses probabilities instead of deterministic rules

2. **Incomplete Data Handling:**
 - Can still predict disease with partial symptoms
3. **Noise Tolerance:**
 - Robust to incorrect or uncertain inputs
4. **Updating Beliefs:**
 - Probabilities updated as new symptoms are observed

Conditional Dependencies Impact

- In Naïve Bayes:
 - Assumes independence → may oversimplify relationships
 - Example: Fever and cough treated independently
- In Bayesian Networks:
 - Captures relationships → more accurate predictions
 - Example: Fever may depend on infection

Conclusion:

- Ignoring dependencies may reduce accuracy
- Modeling dependencies improves realism

Role of Probability Learning

- Probabilities are learned from historical patient data
- Includes:
 - Prior probabilities (disease prevalence)
 - Conditional probabilities (symptom likelihoods)

Learning Methods:

- Maximum Likelihood Estimation (MLE)
- Bayesian Learning (with priors)

Evaluation of Effectiveness

Advantages

- Handles uncertainty effectively
- Works with incomplete data
- Provides probabilistic interpretation
- Scalable to large medical datasets

Challenges (Real-World Constraints)

1. **Data Quality Issues:**
 - Missing or inaccurate patient records

2. **Complex Dependencies:**
 - Difficult to model all relationships
3. **Computational Complexity:**
 - Bayesian networks may be expensive
4. **Ethical Concerns:**
 - Requires careful handling of patient data
5. **Interpretability:**
 - Complex models may be hard for doctors to interpret

Performance Considerations

- Naïve Bayes:
 - Faster but less accurate when dependencies exist
- Bayesian Networks:
 - More accurate but computationally intensive

Conclusion

A probabilistic model combining Bayes' Theorem, Naïve Bayes, and Bayesian Belief Networks provides an effective solution for disease diagnosis under uncertainty. While Naïve Bayes offers simplicity and speed, Bayesian networks improve accuracy by modeling dependencies. The choice of approach depends on data availability, computational resources, and required accuracy.

Result

The designed probabilistic model successfully demonstrates how uncertainty can be managed in medical diagnosis. It highlights that incorporating conditional dependencies and prior knowledge significantly enhances prediction accuracy, making Bayesian approaches suitable for real-world healthcare applications.

2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.

Intelligent Disease Diagnosis using Bayesian Methods

Aim

To design a probabilistic model for disease diagnosis using Bayes' Theorem, Naïve Bayes, and Bayesian Networks.

Bayes' Theorem

Used to compute probability of disease given symptoms:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)} = \frac{0.85(0.20)}{0.25} = 0.68$$

$P(A)$
 $P(A)$
 $P(B | A)$
 $P(B | A)$
 $P(B | \neg A)$
 $P(B | \neg A)$

$P(B)=0.25P(B|A)P(A)=0.17P(A|B)\sim0.68$ Posterior = useful evidence / total evidence

$P(B)=0.25P(B|A)P(A)=0.17P(A|B)\sim0.68$ Posterior = useful evidence / total evidence

- Posterior = Probability of disease after observing symptoms
- Prior = Initial disease probability
- Likelihood = Probability of symptoms given disease

Naïve Bayes Classifier

- Assumes symptoms are independent
- Formula:
 $P(D | S_1...S_n) \propto P(D) \times \prod P(S_i | D)$
- Fast and simple, but may reduce accuracy due to independence assumption

Bayesian Belief Networks

- Uses a DAG to model dependencies between symptoms and diseases
- More accurate as it captures relationships

Handling Uncertainty

- Works with incomplete data
- Updates probabilities as new symptoms appear
- Robust to noise

Key Points

- Naïve Bayes → Fast, less accurate

- Bayesian Networks → Accurate, complex
- Learning from data using MLE or Bayesian methods

Conclusion

Bayesian methods effectively handle uncertainty in diagnosis. Modeling dependencies improves accuracy, making Bayesian networks more reliable for real-world applications.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand